21D009 Networks Project
Modelling Disease Transmission in School Networks: Combining Vaccination and Isolation Strategies for Effective Pandemic Management

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April 5, 2023

Abstract

In recent years, understanding disease transmission has become increasingly important, particularly due to the rapid spread and devastating impact of COVID-19. However, many models used to simulate epidemics and gauge the effectiveness of countermeasures have failed to account for the population's network structure. Traditional epidemic spread models often assume equal transmission capability among connections, while in reality, contact intensity plays a significant role in disease spread. The primary objective of this study is to investigate disease propagation within school networks, the impact of vaccination and isolation measures on decelerating the pandemic, and explore alternative optimal strategies that could be employed instead of resorting to the complete closure of educational institutions.

A modified SEIR model was employed to simulate disease transmission in a primary school network, using a dataset that recorded 125,773 interactions between 232 students and 10 teachers at a primary school in Lyon, France. Findings showed that isolating individuals and their most connected neighbors reduced the maximum number of infected individuals by more than three times, regardless of the contagion rate. Vaccination alone was not sufficient as a standalone strategy, particularly in deeply interconnected communities. The most effective method in slowing down the pandemic was found to be a combination of vaccination and isolation strategies.

1 Introduction

In recent years, the investigation of disease transmission has become increasingly important, mainly due to the rapid spread and devastating impact of COVID-19. However, many models used to simulate epidemics and gauge the effectiveness of countermeasures failed to account for the population's network structure [1].

During the COVID-19 crisis, compartmental models were frequently employed to forecast macroscopic infection and mortality dynamics and evaluate various non-pharmaceutical interventions targeting microscopic person-to-person transmission [**sociologica2019**]. Nonetheless, these models' predictions exhibit high uncertainty and don't allow for precise isolation techniques and measures.

Furthermore, research on smaller scales suggests that random vaccination isn't the most efficient method for addressing disease transmission. Link removal has proven to be a more effective tactic in slowing down epidemics. However, most of the existing literature primarily uses non-weighted approaches, which might not adequately represent disease transmission complexity.

In traditional epidemic spread models, a common assumption is that all connections are equally capable of transmitting infection. In reality, disease spread depends heavily on contact intensity. Hence, the speed of spread and the pathways through which a disease spreads can differ substantially between weighted and non-weighted models [**ploscompbiol2013**]. This implies that the efficacy of various strategies can vary significantly based on the network type. Consequently, it's crucial to consider population network structures when developing epidemic spread models and incorporating prevention measures. This integration will enhance model accuracy and usefulness, ultimately informing more effective public health interventions.

Although obtaining comprehensive social interaction networks is challenging, significant sources such as schools are available. Schools are typically considered major transmission hubs for epidemics, and school closures have significantly slowed down disease spread. However, this approach can lead to adverse effects on students' mental and physical health (e.g., increased anxiety symptoms among school athletes) [3]. In the US, over 17% of children aged 2-8 years have a diagnosed mental, behavioral, or developmental disorder. School psychologists and counselors provide essential support, especially in rural areas where schools might be the sole source of mental health services. In 2017, about 17% of US children lived in food-insecure households. To address this, schools served millions of free and reduced-price meals daily in 2018. Hispanic and Black students, who are more likely to attend schools with over 75% eligibility for free or reduced-price lunch, have now lost access to these resources [**Golberstein2020**].

Therefore, it is crucial to acknowledge that although closing schools can significantly reduce the spread of a pandemic, there are considerable costs associated with this strategy, making it a measure that should only be employed as a final option [7].

Consequently, the primary objective of this study is to delve deeper into understanding how diseases can propagate within school networks, the impact of vaccinations and isolation measures on decelerating the pandemic, and the ways in which various interventions can influence the transmission of the disease. Additionally, the research aims to explore alternative optimal strategies that could be employed instead of resorting to the complete closure of educational institutions.

2 Algorithm

The algorithm employed in this study is a modified version of the SEIR (Susceptible, Exposed, Infectious, and Recovered) model, which also incorporates Vaccinated, Isolated, and Isolated-based-on-neighbors individuals:

- S represents the proportion of susceptible individuals (those able to contract the disease),
- E represents the proportion of exposed individuals (those who have been infected but are not yet infectious),
- I represents the proportion of infected individuals (those capable of transmitting the disease),
- R represents the proportion of recovered individuals (those who have become immune),
- VAC represents the proportion of individuals who have been vaccinated and have a significantly lower probability of getting infected (typically a 95% reduction in probability was used in most modeling scenarios),
- Iso represents the proportion of individuals who have been isolated from the network after falling sick,
- $Iso_neighbors$ represents the proportion of individuals who were isolated for a certain number of days if someone they had communicated with fell sick (the chance of isolation depends on the degree of communication).

The algorithm incorporates several variables into the model:

- Number of initially infected: generally set to 1,
- Basic spread Rate: the Contagiousness of the disease
- External exposure rate: the chance for each node to get sick from outside the system, usually set to 0.25%,
- Vaccination coverage: the number of nodes that were initially vaccinated,
- Vaccination rate: the rate at which individuals were vaccinated in the system (not used for final results, as the amount of data became excessive),

- Weight of the edge: the network was weighted based on the total time of 20s+ interactions, which will be discussed later,
- Chance of detecting a sick person: the probability of isolating a sick individual,
- Isolation time: the time for a node to re-enter the system once isolated,
- Vaccination type: the strategy by which people were vaccinated,
  - Strategy 1 for Vaccination: people were vaccinated randomly in the network,
  - Strategy 2 for Vaccination: people with the highest total number of links were vaccinated,
  - Strategy 3 for Vaccination: people with the highest total weight of connected edges were vaccinated.

Transition probabilities determine all state changes, such as the probability of transmitting a disease from Person 1 to Person 2. This probability is calculated using the following formula:

Probability of transmission = Weight of the interaction (number of 20s intervals) × Basic spread rate

If the result is 0.37, then there is a 37% chance that the person will become sick at the next time interval. To run the simulation, the desired transition probabilities were modified in the $EoN.Gillespie_simple_contagion$ model.

During a time period of 30 intervals, the model was estimated to produce multiple outcomes, such as the maximum number of infected individuals, the number of infected, recovered, isolated, etc. at a given moment t, as well as correlation coefficients between these variables.

3 Data

The dataset, compiled by the SocioPatterns collaboration [**SociopatternsData**], provides a comprehensive contact network acquired through wearable proximity sensors in a primary school. These sensors monitor face-to-face encounters ("contacts") among individuals every 20 seconds. This dynamic contact network records interactions between 232 students and 10 teachers at a primary school in Lyon, France, over two school days (October 1st and 2nd, 2009). The school is comprised of five grades, each having two classes, adding up to 10 classes in total. Each contact event is individually identified, with the start and end times determined accurately within 20-second intervals.

The data encompasses 125,773 interactions between individuals. The interactions for all participants were subsequently summarized for each node (individual), with the sum representing the total number of 20-second or longer interactions between individuals. This sum corresponds to the 2,472 resulting edges for the graph.resulting edges for the graph.

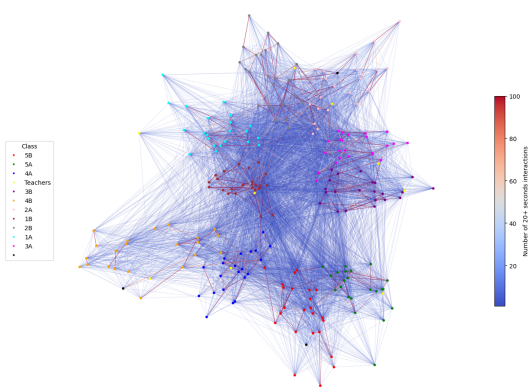

Figure 1: Network with all interactions between individuals

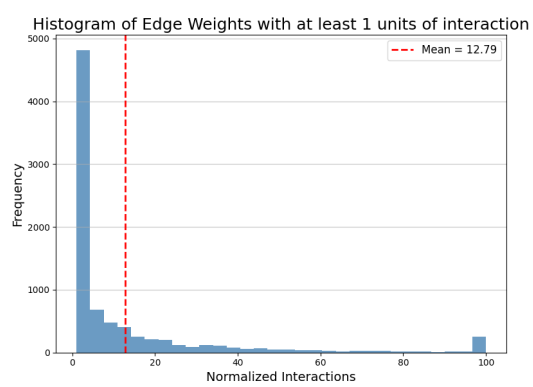

Figure 2: Distribution of intensity of interactions

Figure 3: Interactions between individuals

First, an upper limit on the number of interactions between individuals was introduced, such that if a person had over 100 interactions, it was rounded down to 100. As demonstrated in Figure 8 and 9, the majority of edges between individuals only have one unit of intensity, indicating that most interactions are brief and incidental. Despite using a weighted approach, it was still decided that the more optimal choice would be to set a minimum boundary on interactions.

	Nodes	Edges	Mean Degree	Max Degree	Diameter	Triangles	Clustering	Mean EC	Max EC	Density	EC Range	Avg Edge Weight	Min Sum Edge Weights per Node	Max Sum Edge Weights per Node	Avg Sum Edge Weights per Node	Median Sum Edge Weights per Node	Largest CC size
0	242	8317	68.735537	134	3	1037660.0	0.525542	0.058918	0.118128	0.285210	0.105188	15.122400	130	2594	1039.446281	1000.0	242
1	242	5463	45.148760	90	3	440065.0	0.570899	0.057355	0.128755	0.187339	0.119238	22.500275	117	2582	1015.859504	970.5	242
2	242	3582	28.942149	53	4	192510.0	0.627123	0.054640	0.151404	0.120060	0.146316	33.600228	90	2490	972.462810	937.5	242
3	242	2472	20.429752	42	5	95720.0	0.580028	0.052401	0.142577	0.084771	0.141831	44.297130	34	2430	915.198347	881.5	242
4	242	1938	16.010529	34	7	54720.0	0.554697	0.046045	0.181355	0.066459	0.181341	53.876677	26	2393	862.917355	820.0	242

Figure 4: Descriptive Statistics of the Networks with minimal number of interactions correspond to 1, 2, 5, 10 and 15 time units

In Figure 19, we can observe the descriptive statistics of the networks. As demonstrated, by introducing a minimum weight for nodes, we significantly reduce the size and structure of our network. However, at the same time, statistics such as the median weight of edges per node and the average weight of edges for each node do not change substantially. This corresponds to the fact that the most time-consuming interactions generally make up a larger proportion of the total time spent by individuals, and this should not alter the results of our model significantly. Furthermore, as shown, the size of the largest components remains unchanged, preventing a situation where specific individuals or groups become isolated from the rest of the participants. Therefore, in order to model for the future, a network with a minimal number of interactions equal to 10 units was utilized and an upper bound was established at 100.

Figure 5: Network with lower boundary of 10

After analyzing Figure 5, it becomes apparent that most interactions take place among individuals within the same class. However, there are still a few "bridges" - individuals who interact with members of other classes - and teachers who interact with each other. This suggests that implementing class

closures or carefully selecting bridge individuals may yield similar outcomes to those achieved through complete school closures.

4 Modelling

At first it was attempted to model the spread of the disease on a weighted network with various contagious rates, without the implementation of vaccinations or isolation. As we can see, the once we cut down contingency rate by half, the maximum amount of people, who were sick at one single point in time is getting reduced by about 30% and we do have lower number of people who were getting sick in one moment of time.

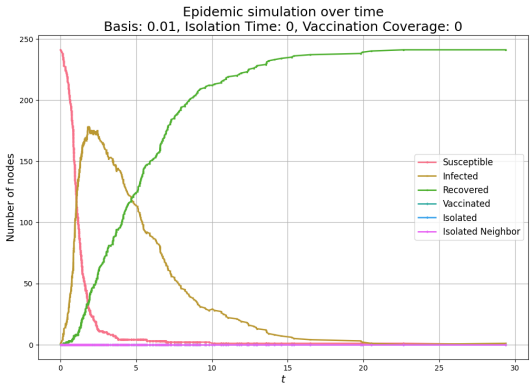

Figure 6: Model With Basic spread rate 0.01

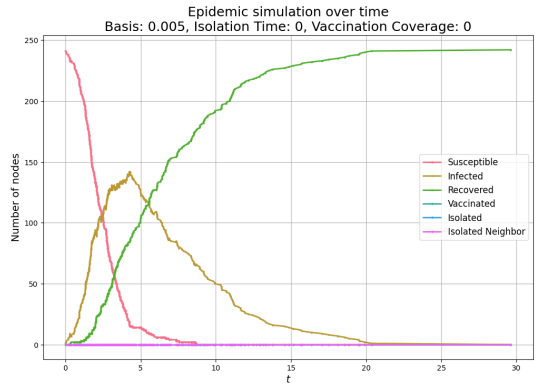

Figure 7: Model With Basic spread rate 0.005

By implementing individual isolation and that of their neighbors (with a 50% probability of identifying the infected person and isolating them, and the neighbor's isolation based on weight/100), the recovery time for a person is established at 4 days. Consequently, there is still a possibility that the infected individual may not be detected during their entire duration as a disease carrier.

This strategy has been remarkably effective in considerably decreasing the peaks in the number of infected individuals at any given moment (reduction of almost 3 times). Furthermore, at the conclusion of the 30-day simulation, some individuals remained unaffected by the disease (approximately 10%), even in scenarios with the highest contagion rates. Additionally, throughout most of the time, the number of isolated cases due to identified disease and the number of isolated neighbors maintain very similar levels.

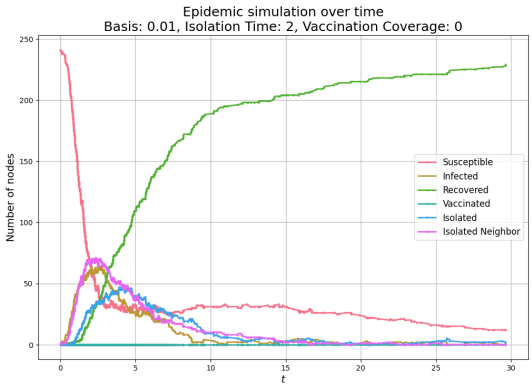

Figure 8: Model With Basic spread rate 0.01 and Isolation for 2 days

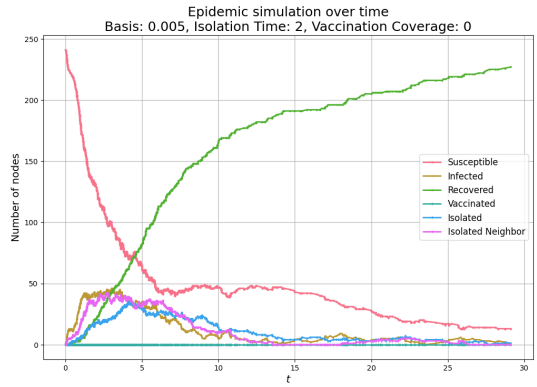

Figure 9: Model With Basic spread rate 0.005 and Isolation for 2 days

Next, three vaccination strategies were attempted to vaccinate 20% of individuals:

1. Randomly vaccinating 20% of the nodes.
2. Vaccinating the top 20% of nodes with a higher number of edges (node degree)
3. Vaccinating the top 20% of nodes based on node weight.

Vaccinated individuals were 95% less likely to get sick and, correspondingly, required a significantly higher intensity of interaction to become ill.

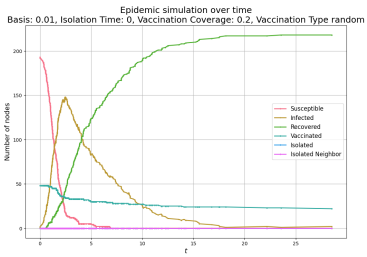

Figure 10: Model With Basic spread rate 0.01, Randomized Vaccination

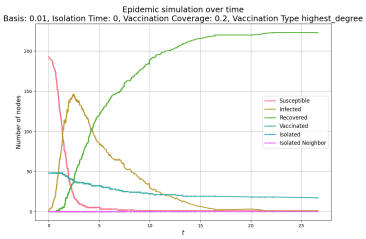

Figure 11: Model With Basic spread rate 0.01, Edge Based Vaccination

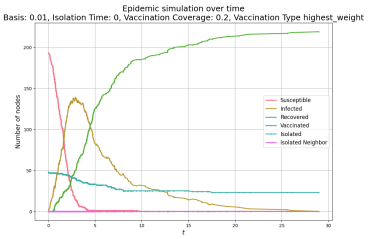

Figure 12: Model With Basic spread rate 0.01, Weight Based Vaccination

We can observe that vaccinating a portion of the population helps to both reduce the peak number of sick individuals and delay the moment the peak occurs. However, this is primarily due to vaccinated individuals being less likely to get sick. The application of weight-based and edge-based vaccination techniques demonstrated some positive results in reducing the peak values compared to random assignment, but the effect was not substantial. It's worth noting that some modifications to the vaccination modeling could be introduced, as highly central individuals, such as teachers, were still getting sick relatively quickly even if they were vaccinated.

Finally, it was decided to combine both isolation techniques and partial vaccination (weighted vaccination was chosen).

Figures 15 and 16 demonstrate that when incorporating both weight-based vaccination and isolation techniques, we substantially reduce the spikes in the maximum number of sick people and the total size of the epidemic. It is also important to note that, at this point, most of the spread occurs not due to network transmission, but rather because of the external chance of contracting the disease, which is equal to 0.25% for each unvaccinated individual.

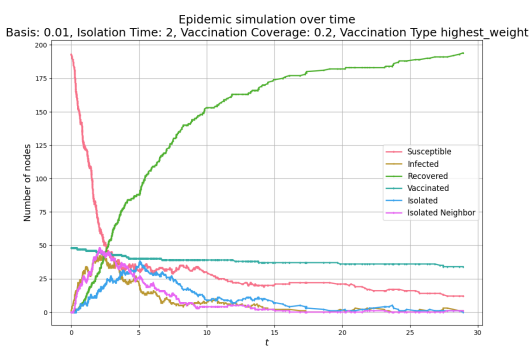

Figure 13: Model With Basic spread rate 0.01 and Isolation for 2 days, and 20% vaccinations

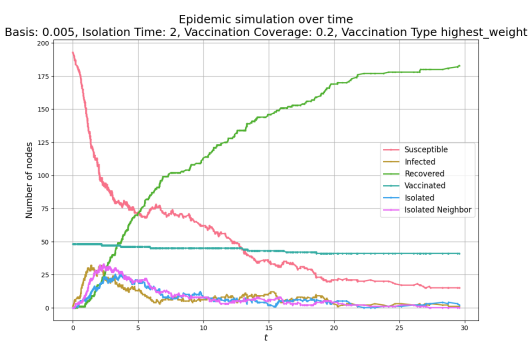

Figure 14: Model With Basic spread rate 0.005 and Isolation for 2 days, and 20% vaccinations

To obtain more consistent results, 50 simulations for each type of network was attempted, the summarized results for all models can be found in attachments.

The results for people who were vaccinated by different methods was very similar for the most of the models, however, however we do see that the spike size of maximum number of infected was slightly smaller on average for weighted approach and it has slightly less sick and isolated individuals by day 10.

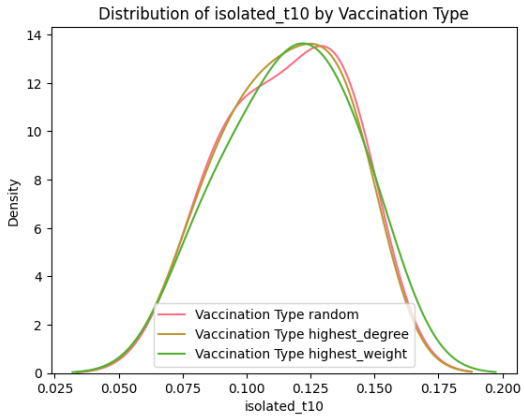

Figure 15: Distribution of isolated by day 10 for each vaccination type

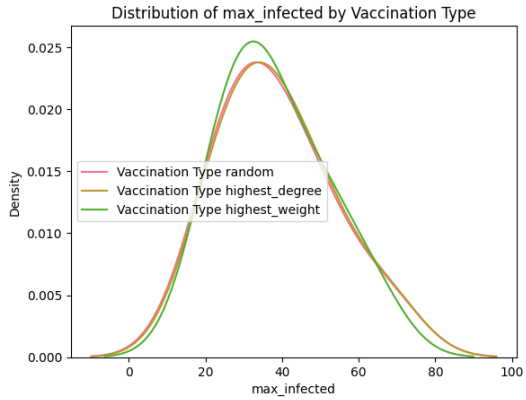

Figure 16: Distribution of Maximum infected for each vaccination type

Upon examining the network states on day 10, it becomes apparent that the middle of the network contains a higher proportion of vaccinated individuals. However, most of their neighbors had already contracted the disease by this point. As a result, the partial vaccination did not significantly impede the spread of the disease throughout the network. This was due to the fact that once potential carriers had come into contact with the vaccinated individuals' neighbors, they had already had enough exposure to other potential carriers. This approach did, however, protect the vaccinated individuals from contracting the disease.

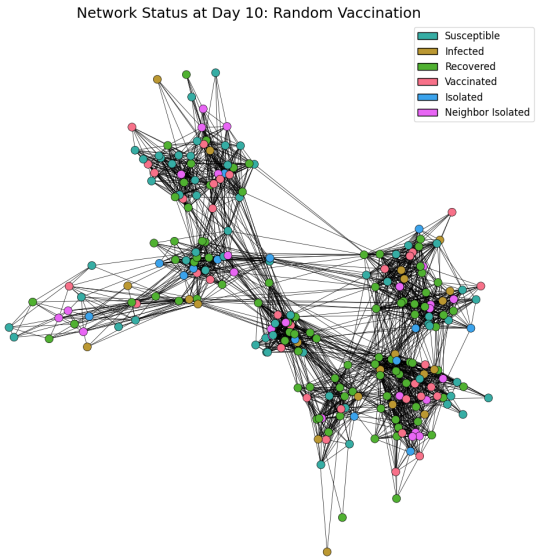

Figure 17: Node statues By day 10 for Random Vaccination Assignment

Overall, we assume that even if a vaccine is not 100% effective, targeting more connected individuals or tightly connected communities will not substantially slow down the spread of the disease within the network itself. An OLS regression was also conducted on the dependent variable "Max infected"

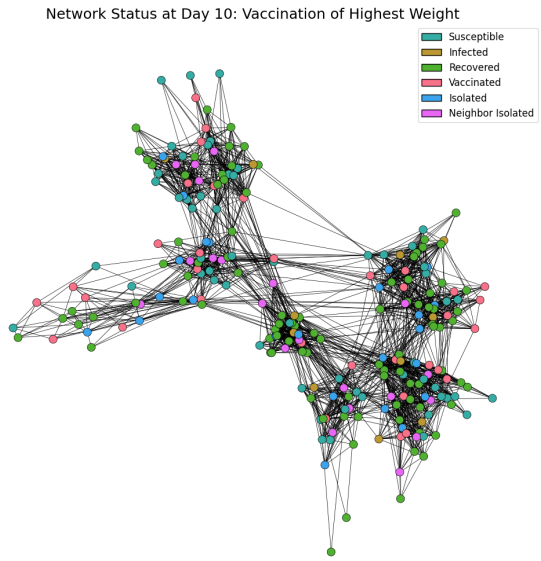

Figure 18: Node statues By day 10 for Weight Based Vaccination Assignment

to determine the significance of various factors. The results showed that vaccination type was not significant (with a p-value of 0.93) when comparing random and weighted assignment. However, vaccination coverage, isolation time, and isolation rate were all significant at a 1% significance level.

5 Conclusion

Throughout the simulation process, several notable results were obtained. The isolation of individuals and their most connected neighbors was found to be highly effective in slowing down the pandemic. Assuming that the measure of class isolation is sufficiently accurate, this approach demonstrated a reduction in the maximum number of infected individuals by more than three times, regardless of the contagion rate.

While vaccination did help to decelerate the pandemic and resulted in lower peak infection numbers, it was not sufficient as a standalone strategy, especially in deeply interconnected communities. Targeted vaccination provided some improvement in slowing pandemic rates, but its impact was limited in tightly linked communities.

The most effective method in slowing down the pandemic was found to be a combination of vaccination and isolation strategies. It is important to note that further model refinements could be introduced, such as examining additional network structures and employing different model formulations, which may yield different results.

References

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Attachments

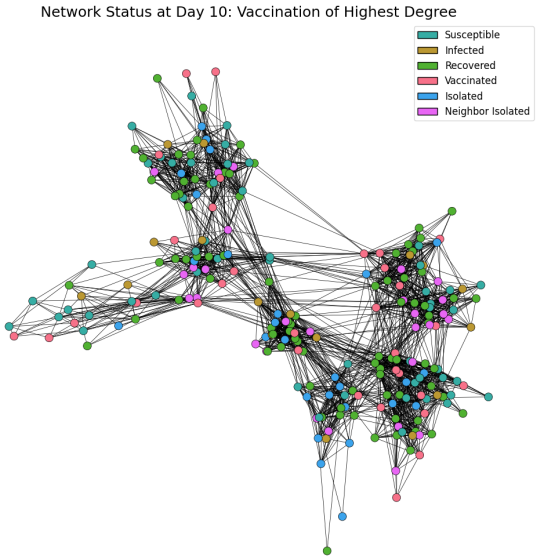

Figure 19: Node statues By day 10 for Degree Based Vaccination Assignment

vaccination_coverage	0	0	0	0	0	0	0	0	0	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
vaccination_type	0	0	0	1	1	1	2	2	2	0	0	0	1	1	1	2	2
isolation_time	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
basis	0.0025	0.005	0.01	0.0025	0.005	0.01	0.0025	0.005	0.01	0.0025	0.005	0.01	0.0025	0.005	0.01	0.0025	0.005
max_infected	34.9667	49.2	67.0667	35.6667	49.7	67.2	34.2333	50.6333	63.3333	26.1333	37.3333	50.6	27.7333	37.7333	50.7333	27.9333	36.2
t	29.8096	29.7909	29.7766	29.8055	29.7352	29.8106	29.8551	29.7743	29.7523	29.8142	29.7475	29.8281	29.7791	29.7558	29.7567	29.7639	29.7318
infected	0.0081	0.0084	0.007	0.0101	0.008	0.0069	0.01	0.0101	0.0069	0.0077	0.0077	0.007	0.0073	0.0065	0.0052	0.0081	0.0112
recovered	0.821	0.850	0.888	0.821	0.851	0.877	0.823	0.851	0.877	0.672	0.707	0.744	0.675	0.705	0.749	0.676	0.698
susceptible	0.123	0.096	0.066	0.117	0.099	0.072	0.120	0.095	0.075	0.108	0.078	0.065	0.108	0.090	0.063	0.099	0.082
exposed	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.174	0.165	0.149	0.172	0.166	0.147	0.175	0.167
isolated	0.037	0.031	0.022	0.040	0.028	0.029	0.037	0.031	0.027	0.029	0.031	0.025	0.031	0.025	0.024	0.034	0.031
healthy_isolated	0.011	0.015	0.017	0.013	0.015	0.015	0.010	0.012	0.014	0.009	0.011	0.009	0.007	0.008	0.011	0.008	0.011
infected_t10	0.114	0.174	0.238	0.111	0.168	0.241	0.107	0.169	0.221	0.079	0.124	0.179	0.086	0.119	0.178	0.080	0.119
recovered_t10	0.143	0.124	0.104	0.146	0.126	0.106	0.151	0.116	0.108	0.125	0.111	0.095	0.124	0.111	0.098	0.126	0.107
susceptible_t10	0.445	0.303	0.185	0.442	0.302	0.174	0.462	0.304	0.179	0.390	0.280	0.166	0.384	0.290	0.169	0.387	0.286
exposed_t10	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.193	0.191	0.183	0.192	0.188	0.184	0.193	0.188
isolated_t10	0.137	0.143	0.140	0.137	0.144	0.138	0.131	0.151	0.144	0.107	0.121	0.124	0.107	0.119	0.122	0.110	0.122
healthy_isolated_t10	0.161	0.256	0.332	0.164	0.260	0.340	0.150	0.260	0.349	0.106	0.175	0.253	0.107	0.173	0.249	0.105	0.178
simulation	14.500	14.500	14.500	14.500	14.500	14.500	14.500	14.500	14.500	14.500	14.500	14.500	14.500	14.500	14.500	14.500	14.500

vaccination_coverage	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
vaccination_type	0	0	1	1	1	2	2	2	0	0	0	1	1	1	2	2	2
isolation_time	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
basis	0.005	0.01	0.0025	0.005	0.01	0.0025	0.005	0.01	0.0025	0.005	0.01	0.0025	0.005	0.01	0.0025	0.005	0.01
max_infected	37.3333	50.6	27.7333	37.7333	50.7333	27.9333	36.2	48.9667	18.9667	44983	36.0333	19.3667	25.3667	36.5667	20.3333	25.3333	36.3667
t	29.7475	29.8281	29.7791	29.7558	29.7567	29.7639	29.7318	29.7871	29.7855	29.7348	29.701	29.7698	29.7956	29.7784	29.8062	29.7789	29.6597
infected	0.0077	0.007	0.0073	0.0065	0.0052	0.0081	0.0112	0.0087	0.0099	0.0073	0.0058	0.0083	0.0091	0.007	0.0101	0.0084	0.0048
recovered	0.707	0.744	0.675	0.705	0.749	0.676	0.698	0.749	0.515	0.554	0.604	0.514	0.554	0.601	0.519	0.552	0.611
susceptible	0.078	0.065	0.108	0.090	0.063	0.099	0.082	0.058	0.087	0.067	0.053	0.085	0.066	0.052	0.083	0.068	0.049
exposed	0.165	0.149	0.172	0.166	0.147	0.175	0.167	0.149	0.352	0.342	0.311	0.357	0.337	0.311	0.352	0.340	0.302
isolated	0.031	0.025	0.031	0.025	0.024	0.034	0.031	0.023	0.030	0.023	0.021	0.030	0.025	0.021	0.029	0.026	0.024
healthy_isolated	0.011	0.009	0.007	0.008	0.011	0.008	0.011	0.012	0.006	0.006	0.006	0.005	0.008	0.007	0.007	0.006	0.009
infected_t10	0.124	0.179	0.086	0.119	0.178	0.080	0.119	0.170	0.057	0.083	0.116	0.053	0.079	0.121	0.056	0.071	0.122
recovered_t10	0.111	0.095	0.124	0.111	0.098	0.126	0.107	0.100	0.104	0.093	0.088	0.105	0.097	0.087	0.107	0.097	0.085
susceptible_t10	0.280	0.166	0.384	0.290	0.169	0.387	0.286	0.171	0.315	0.237	0.160	0.323	0.245	0.151	0.317	0.254	0.155
exposed_t10	0.191	0.183	0.192	0.188	0.184	0.193	0.188	0.187	0.385	0.381	0.376	0.385	0.379	0.372	0.384	0.383	0.367
isolated_t10	0.121	0.124	0.107	0.119	0.122	0.110	0.122	0.124	0.079	0.093	0.097	0.078	0.091	0.100	0.078	0.091	0.103
healthy_isolated_t10	0.175	0.253	0.107	0.173	0.249	0.105	0.178	0.248	0.060	0.113	0.164	0.057	0.109	0.169	0.058	0.104	0.167
simulation	14.500	14.500	14.500	14.500	14.500	14.500	14.500	14.500	14.500	14.500	14.500	14.500	14.500	14.500	14.500	14.500	14.500

Figure 20: Table of all models

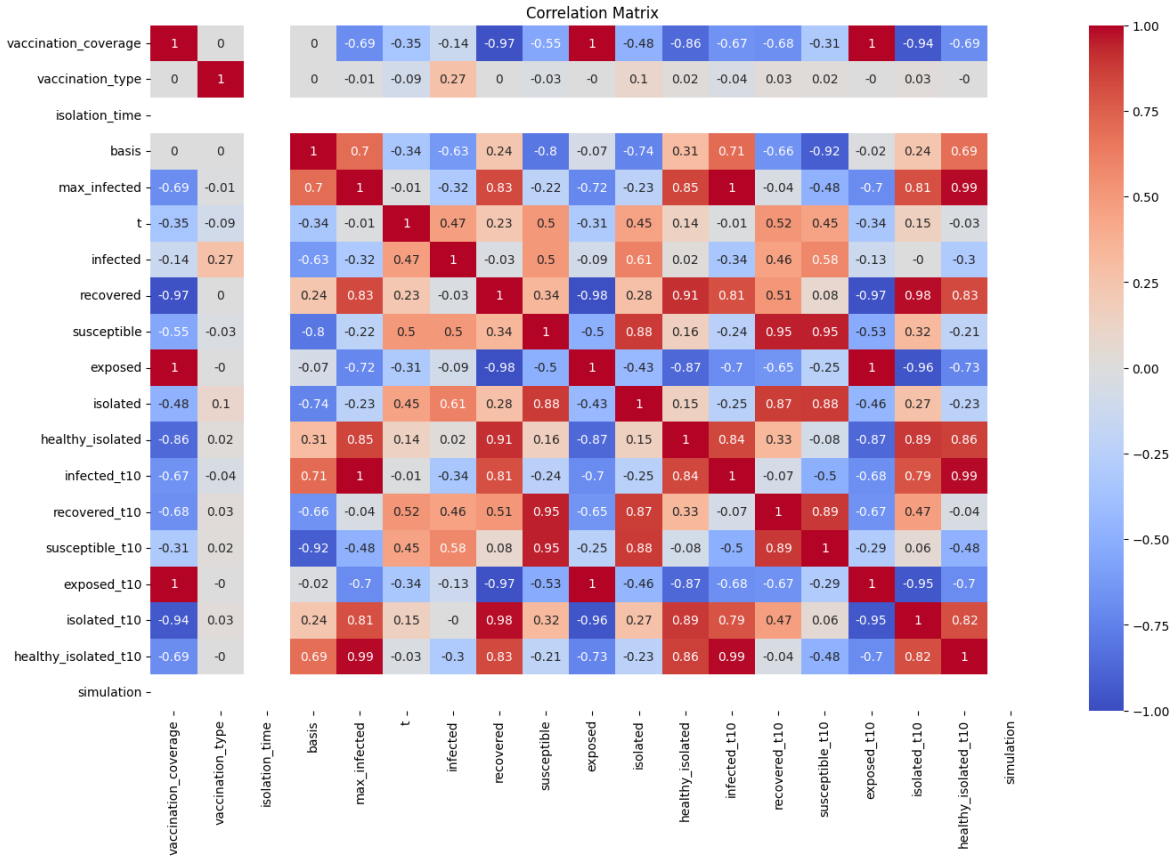

Figure 21: All Correlations

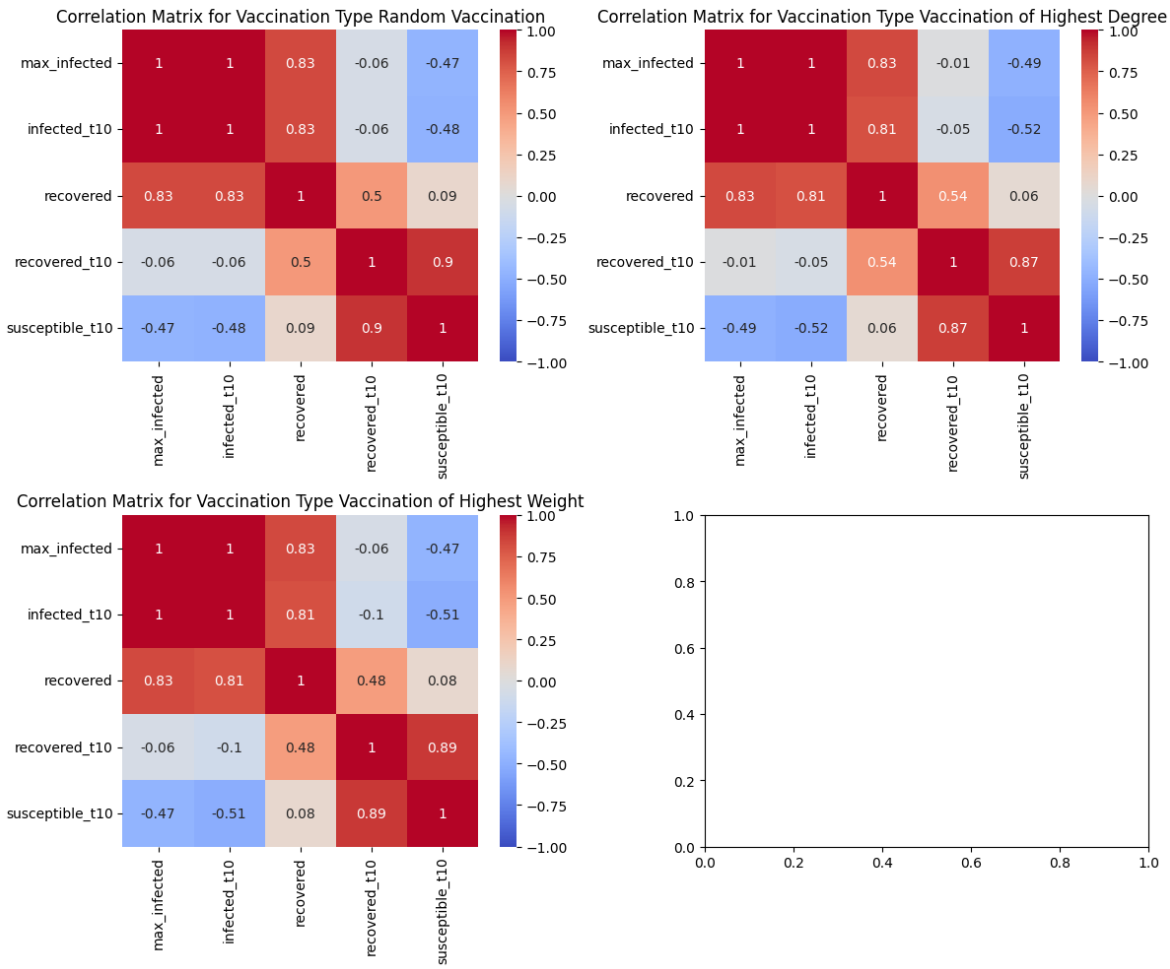

Figure 22: Correlations Vaccination type